CHARACTERIZATIONS OF UNIVERSAL CLOSEDNESS OF A REPRESENTABLE MORPHISM OF DECENT SPACES

THE STACKS PROJECT AUTHORS

Editorial note. No omission markers in-span.

Allowed standard prerequisites (editorial).

- Specialization lifting. lemma-specialization (carded L26708). Lifts specializations through the etale cover.
- Scheme case. Morphisms, lemma universally-closed-characterization (cross-chapter, named). The scheme case for U -> Y.
- Valuative criterion points. Morphisms of Spaces, lemma valuative-criterion (cross-chapter, named). The spectra of valuation rings in the statement.

Source reference. Original reference 00IA: focused-seeds/geometry/S9-decent-characterize-universally-closed/ref-tags.txt, source lines 5–5. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original lemma 03KF: sources/stacks-decent-spaces/pinned-original-source.tex, source lines 1126–1126. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original lemma 03K2: sources/stacks-decent-spaces/pinned-original-source.tex, source lines 1544–1544. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original reference 03IX: focused-seeds/geometry/S9-decent-characterize-universally-closed/ref-tags.txt, source lines 6–6. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original reference 03HF: focused-seeds/geometry/S9-decent-characterize-universally-closed/ref-tags.txt, source lines 7–7. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original reference 03KA: focused-seeds/geometry/S9-decent-characterize-universally-closed/ref-tags.txt, source lines 8–8. Navigation pointer only: the counted

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proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Proposition 0.1. *Let S be a scheme. Let $f : X \to Y$ be a morphism of algebraic spaces over S. Assume f is quasi-compact, and X is decent. Then f is universally closed if and only if the existence part of the valuative criterion holds.*

Proof. In Morphisms of Spaces, Lemma 03KA we have seen one of the implications. To prove the other, assume that f is universally closed. Let

$$\begin{array}{ccc} \mathrm{Spec}(K) & \longrightarrow & X \\ \downarrow & & \downarrow \\ \mathrm{Spec}(A) & \longrightarrow & Y \end{array}$$

be a diagram as in Morphisms of Spaces, Definition 03IX. Let $X_A = \mathrm{Spec}(A) \times_Y X$, so that we have

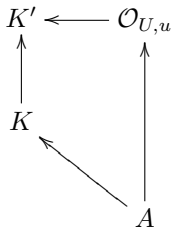
$$\begin{array}{ccc} \mathrm{Spec}(K) & \longrightarrow & X_A \\ & \searrow & \downarrow \\ & & \mathrm{Spec}(A) \end{array}$$

By Morphisms of Spaces, Lemma 03HF we see that $X_A \to \mathrm{Spec}(A)$ is quasi-compact. Since $X_A \to X$ is representable, we see that X_A is decent also, see Lemma 03KF. Moreover, as f is universally closed, we see that $X_A \to \mathrm{Spec}(A)$ is universally closed. Hence we may and do replace X by X_A and Y by $\mathrm{Spec}(A)$.

Let $x' \in |X|$ be the equivalence class of $\mathrm{Spec}(K) \to X$. Let $y \in |Y| = |\mathrm{Spec}(A)|$ be the closed point. Set $y' = f(x')$; it is the generic point of $\mathrm{Spec}(A)$. Since f is universally closed we see that $f(\overline{\{x'\}})$ contains $\overline{\{y'\}}$, and hence contains y. Let $x \in \overline{\{x'\}}$ be a point such that $f(x) = y$. Let U be a scheme, and $\varphi : U \to X$ an étale morphism such that there exists a $u \in U$ with $\varphi(u) = x$. By Lemma 03K2 and our assumption that X is decent there exists a specialization $u' \rightsquigarrow u$ on U with $\varphi(u') = x'$. This means that there exists a common field extension $K \subset K' \supset \kappa(u')$ such that

$$\begin{array}{ccc} \mathrm{Spec}(K') & \longrightarrow & U \\ \downarrow & & \downarrow \\ \mathrm{Spec}(K) & \longrightarrow & X \\ & \searrow & \downarrow \\ & & \mathrm{Spec}(A) \end{array}$$

is commutative. This gives the following commutative diagram of rings



By Algebra, Lemma 00IA we can find a valuation ring $A' \subset K'$ dominating the image of $\mathcal{O}_{U,u}$ in K'. Since by construction $\mathcal{O}_{U,u}$ dominates A we see that A' dominates A also. Hence we obtain a diagram resembling the second diagram of Morphisms of Spaces, Definition 03IX and the proposition is proved. □